

Total No. of Questions : 10]

SEAT No. :

**P2411**

**[4758]-579**

[Total No. of Pages : 2

**T.E. (Computer Engineering)**  
**OPERATING SYSTEMS DESIGN**

**(2012 Course) (End - Sem.) (Semester - I) (310242)**

*Time : 2.30 Hours]*

*[Max. Marks : 70*

*Instructions to the candidates:*

- 1) *All questions are compulsory.*
- 2) *Figures to the right indicate full marks.*

**Q1) a)** Explain in short - BIOS, MBR and init( ) process. **[6]**

b) What is kernel? What facility kernel should provide? **[4]**

OR

**Q2) a)** What are different multithreading models? **[6]**

b) What is TLB? why it is used? **[4]**

**Q3) a)** Explain following algorithms of file management. **[6]**

i) iget ii) iput

b) Give the details of Uarea field. **[4]**

OR

**Q4) a)** Why is the principle of locality crucial to the use of virtual memory? Explain with example. **[4]**

b) If the page address stream is {2, 3, 2, 1, 5, 2, 4, 5, 3, 2, 5, 2}, and frame size is 3. Identify the page faults occurred using FIFO, LRU. **[6]**

**P.T.O.**

- Q5)** a) What are the problems in multiprocessor systems? provide solutions to overcome them. [8]
- b) Explain IPC mechanisms used in System V. [8]

OR

- Q6)** a) What is process tracing? Mention its advantages and disadvantages. [8]
- b) Explain in short - pipe, semaphore, signal and mutex. [8]
- Q7)** a) What is AWK scripting? Write an AWK script to print squares of numbers from 1 to 10. [8]
- b) What is secure boot? State the difference between BIOS and UEFI. [8]

OR

- Q8)** a) What is grep Utility? What are the grep variations? Explain with example. [8]
- b) What is make utility? Explain it with example. Consider your own make file. [8]
- Q9)** a) Enlist different characteristics of real time system and explain. [6]
- b) Explain static priority-driven preemptive approach for real time scheduling. [6]
- c) Compare Hard, soft and Firm real time systems. [6]

OR

- Q10)** a) Explain data structures used in Linux scheduling. [6]
- b) Write short note on frame of references for handheld system. [6]
- c) Compare Windows NTFS and ReFS file systems. [6]

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